ASSIGNMENT 8

BASIC SQL QUERIES AND AGGREGATE FUNCTIONS

Project: Library Management System | Database: LibraryManagementSystem

Student Name: _____ **Class:** _____

1. Introduction

SQL (Structured Query Language) is used to retrieve and manage information in a relational database. This assignment demonstrates filtering, sorting, grouping and aggregate functions using the LibraryManagementSystem database.

Sample-data note: The outputs in this report are illustrative expected results based on a consistent example dataset: 5 book records (total copies 5, 4, 3, 6 and 5; category counts 1, 3 and 1), 4 members, 2 staff members, 6 loans (2 per first two members, 1 per next two members), 3 reservations and 2 fines (■20.00 and ■15.00). Run the queries against matching inserted data to reproduce these outputs. Dates and names are example data.

2. Filtering Records Using WHERE

Query 1: Display books with fewer than three available copies.

```
SELECT BookID, Title, AvailableCopies
FROM Books
WHERE AvailableCopies < 3;
```

BookID	Title	AvailableCopies
102	Python Programming	2
103	Artificial Intelligence	1

Explanation: WHERE filters the Books table and returns only records whose AvailableCopies value is less than 3.

3. Sorting Records Using ORDER BY

Query 2: Display books in descending order of total copies.

```
SELECT BookID, Title, TotalCopies
FROM Books
ORDER BY TotalCopies DESC, BookID ASC;
```

BookID	Title	TotalCopies
104	Computer Networks	6
101	Database Management Systems	5
105	Java Programming	5
102	Python Programming	4
103	Artificial Intelligence	3

Explanation: ORDER BY sorts the records by TotalCopies from highest to lowest. BookID is used as a secondary sort for ties.

4. Grouping Records Using GROUP BY

Query 3: Count the number of books in each category.

```
SELECT CategoryID, COUNT(*) AS TotalBooks
```

```
FROM Books
GROUP BY CategoryID
ORDER BY CategoryID;
```

CategoryID	TotalBooks
1	1
2	3
3	1

Explanation: GROUP BY places books with the same CategoryID into groups. COUNT() returns the number of book records in each group.

5. Filtering Groups Using HAVING

Query 4: Display categories containing more than one book.

```
SELECT CategoryID, COUNT(*) AS TotalBooks
FROM Books
GROUP BY CategoryID
HAVING COUNT(*) > 1;
```

CategoryID	TotalBooks
2	3

Explanation: HAVING filters grouped results. Only categories with more than one book are displayed.

6. Aggregate Functions

Aggregate functions perform calculations on a set of records and return a single value.

Query 5: COUNT() – Count all book records.

```
SELECT COUNT(*) AS TotalBooks
FROM Books;
```

TotalBooks
5

Explanation: COUNT(*) returns the number of rows in Books.

Query 6: SUM() – Calculate the total number of copies.

```
SELECT SUM(TotalCopies) AS TotalCopies
FROM Books;
```

TotalCopies
23

Explanation: SUM() adds the TotalCopies values across all book records.

Query 7: AVG() – Calculate the average number of copies per book record.

```
SELECT AVG(TotalCopies) AS AverageCopies
FROM Books;
```

AverageCopies
4.6000

Explanation: AVG() calculates the arithmetic average of TotalCopies.

Query 8: MIN() – Find the smallest total-copy count.

```
SELECT MIN(TotalCopies) AS MinimumCopies
FROM Books;
```

MinimumCopies
3

Explanation: MIN() returns the smallest TotalCopies value.

Query 9: MAX() – Find the largest total-copy count.

```
SELECT MAX(TotalCopies) AS MaximumCopies
FROM Books;
```

MaximumCopies
6

Explanation: MAX() returns the largest TotalCopies value.

7. Additional Aggregate Queries

Query 10: Count loans according to their status.

```
SELECT Status, COUNT(*) AS TotalLoans
FROM Loans
GROUP BY Status
ORDER BY Status;
```

Status	TotalLoans
Issued	3
Returned	3

Explanation: The query groups loan records by Status and counts the records in each group.

Query 11: Calculate fine statistics.

```
SELECT
    COUNT(*) AS TotalFines,
    SUM(Amount) AS TotalFineAmount,
    AVG(Amount) AS AverageFine,
    MIN(Amount) AS MinimumFine,
    MAX(Amount) AS MaximumFine
FROM Fines;
```

TotalFines	TotalFineAmount	AverageFine	MinimumFine	MaximumFine
2	35.00	17.50	15.00	20.00

Explanation: This query uses COUNT(), SUM(), AVG(), MIN() and MAX() to summarize the fine records.

8. Conclusion

The WHERE, ORDER BY, GROUP BY and HAVING clauses help filter, sort and group data. Aggregate functions such as COUNT(), SUM(), AVG(), MIN() and MAX() provide useful summaries. These SQL features make it easier to retrieve and analyse information in the Library Management System.